MULTI-SENSOR BORDER TREAT DETECTION & CLASSIFICATION SYSTEM

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INTRODUCTION

1. Border surveillance is a critical requirement in modern defence systems
2. Threats can enter through:
3. Land (human intruders, vehicles)
4. Air (drones, UAVs)
5. Traditional single-sensor systems:
6. Generate false alarms
7. Fail to classify the type of intrusion
8. This project proposes a multi-sensor intelligent detection system

PROBLEM STATEMENT

1. Detect intrusion in border areas effectively
2. Distinguish land-based and air-based threats
3. Reduce false alarms due to:
4. Birds
5. Animals
6. Environmental noise
7. Design a low-cost, scalable, student-level prototype

OBJECTIVES

Detect intrusion through land and air routes

Classify land intrusion as:

1. Human movement
2. Vehicle movement

Classify air intrusion as:

1. Drone
2. Bird / safe object

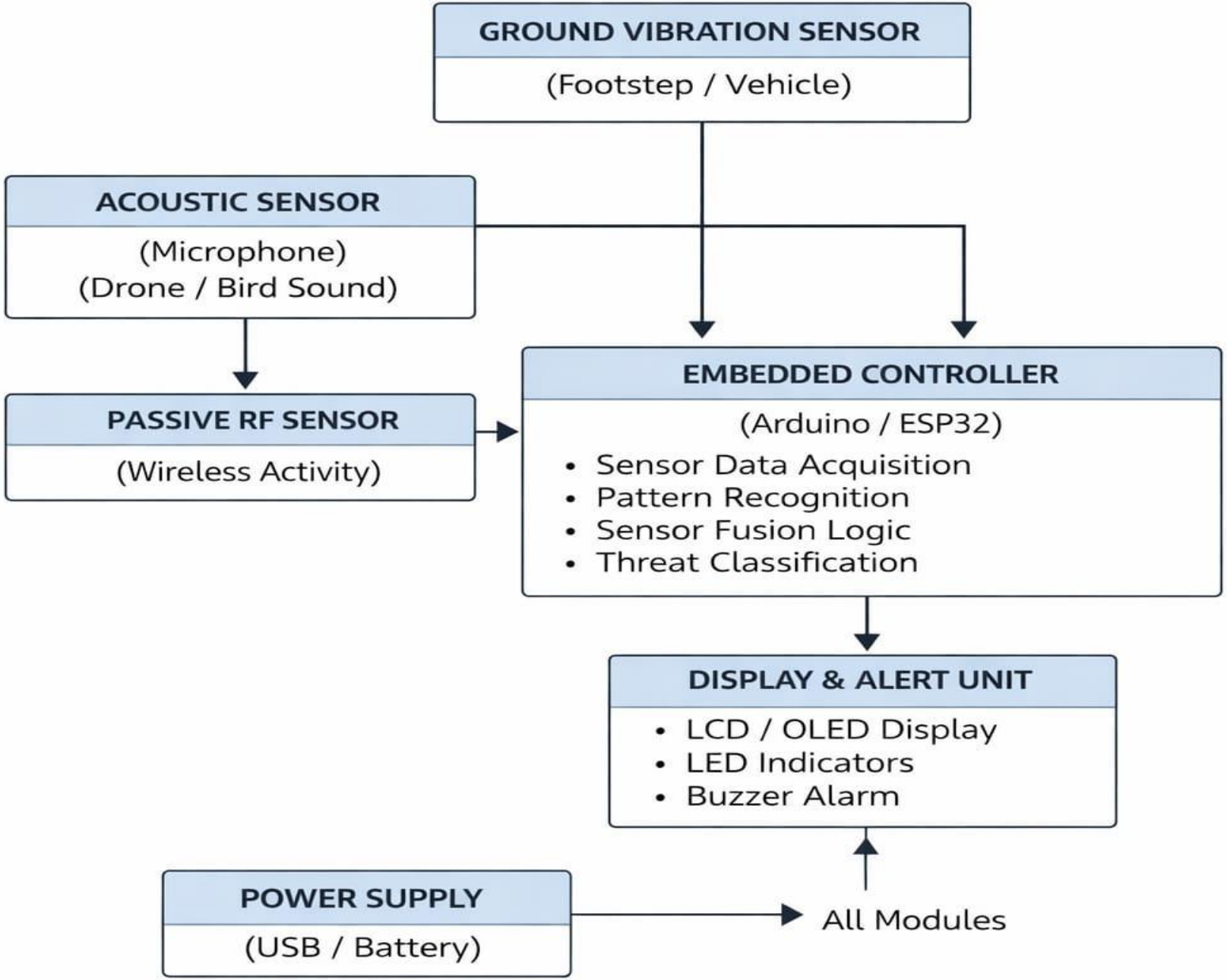
Implement sensor fusion for reliable decision making

provide real-time alerts with threat level indication

SYSTEM OVERVIEW

The proposed system consists of:

- Ground vibration sensing unit
- Acoustic sensing unit
- Passive RF sensing unit
- Embedded controller(Arduino/ESP32)
- Display and alert unit
- The system functions as a border surveillance node.


GROUND VIBRATION SENSOR
(Footstep / Vehicle)
ACOUSTIC SENSOR
(Microphone)
(Drone / Bird Sound)
PASSIVE RF SENSOR
(Wireless Activity)
EMBEDDED CONTROLLER
(Arduino / ESP32)
• Sensor Data Acquisition
• Pattern Recognition
• Sensor Fusion Logic
• Threat Classification
DISPLAY & ALERT UNIT
• LCD / OLED Display
• LED Indicators
• Buzzer Alarm
POWER SUPPLY
(USB / Battery)
All Modules

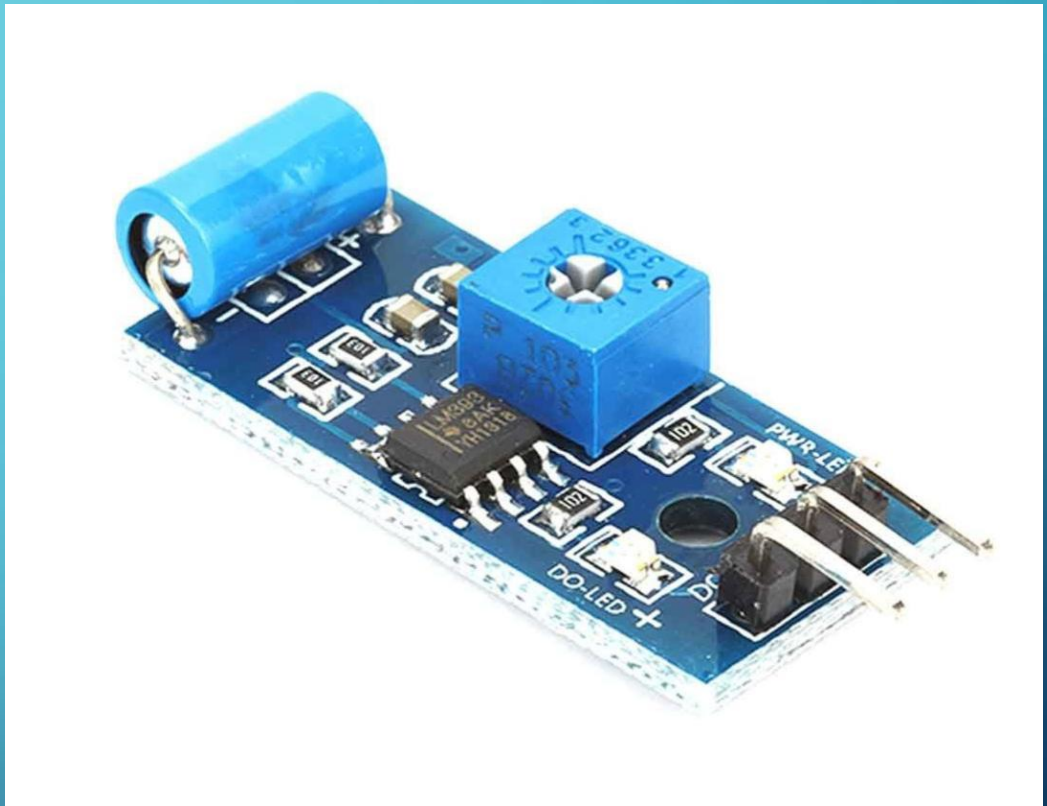
GROUND VIBRATION SENSOR (LAND DETECTION)

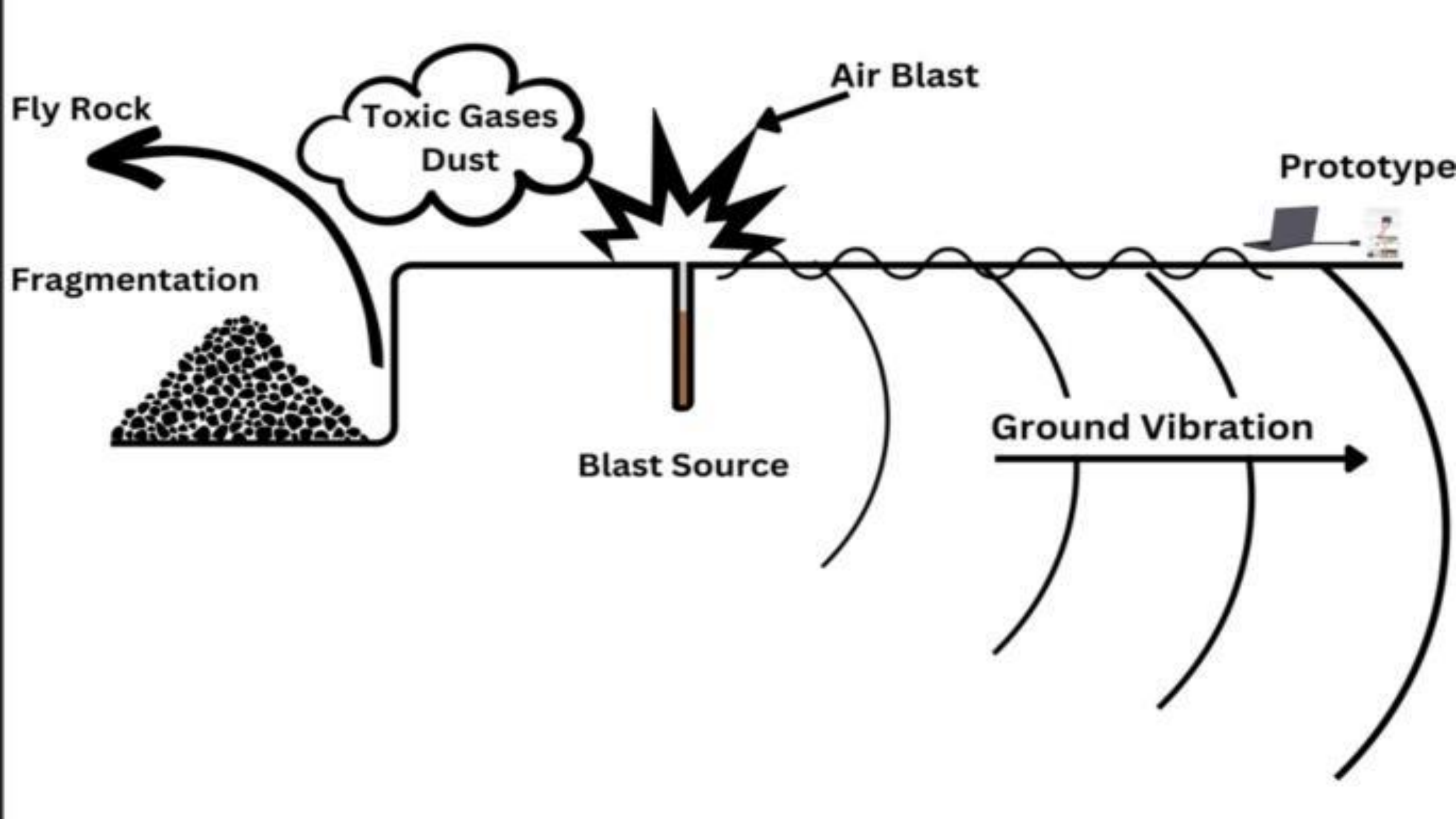
- Detects mechanical vibrations on the ground
- Used to identify land-based intrusions

Classification logic:

- Low amplitude periodic vibrations→Human footsteps
- High-amplitude continuous vibrations→Vehicle movement

Output:LAND:HUMANLAND:VEHICLE



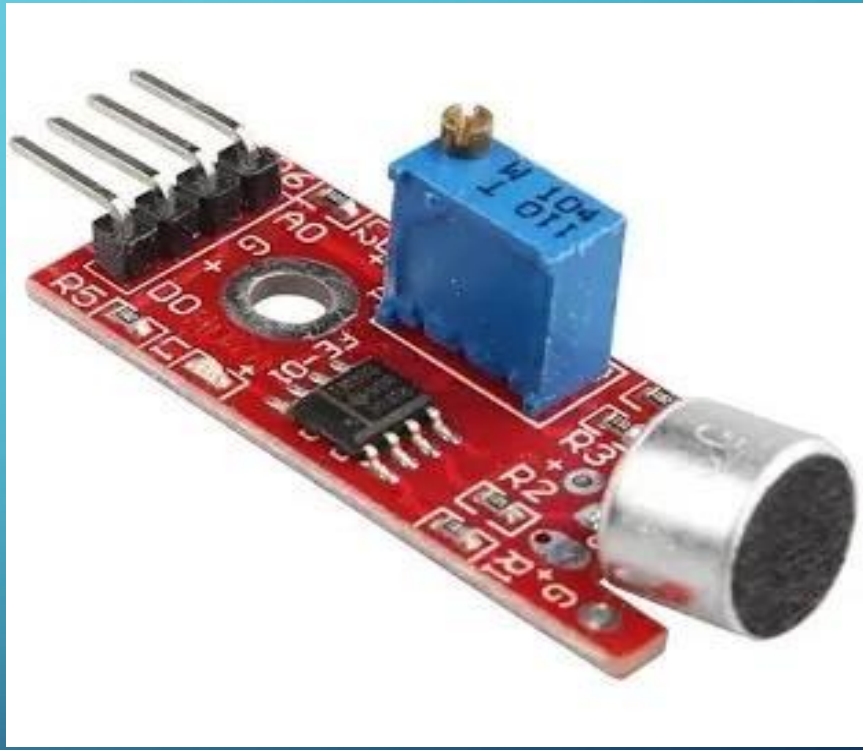

Fly Rock
Toxic Gases
Dust
Air Blast
Prototype
Fragmentation
Ground Vibration
Blast Source

ACOUSTIC SENSOR (DRONE/BIRD DETECTION)

- Uses a microphone to monitor sound patterns
- Drones generate steady buzzing sounds
- Birds produce irregular, short-duration chirps

Sound based Classification:

1. Continuous stable sound→Drone-like
2. Randompeaks→Bird/noise

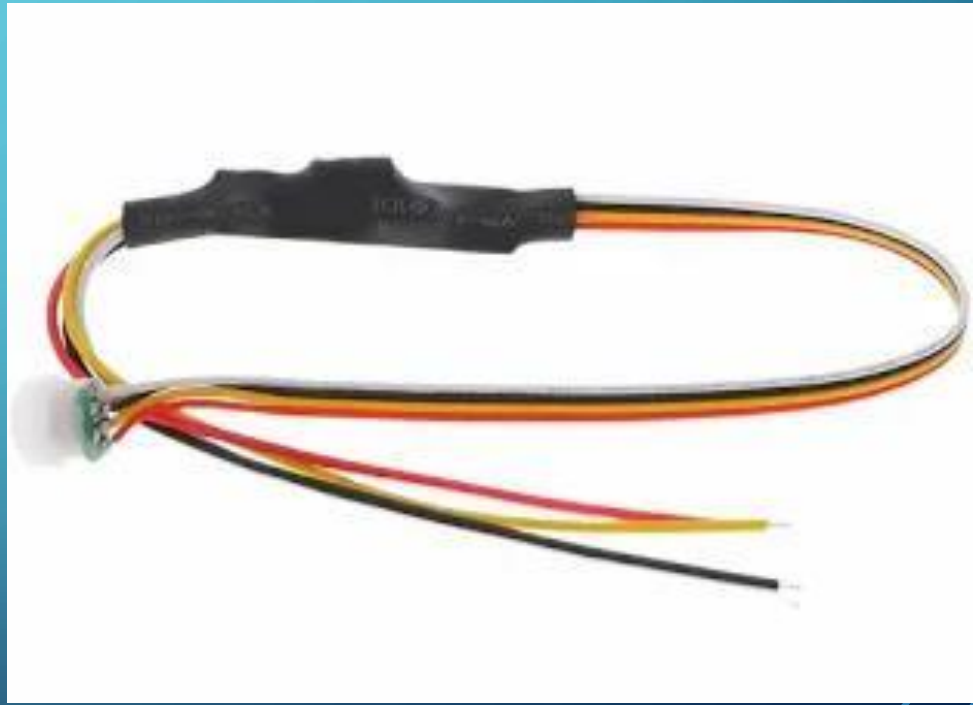


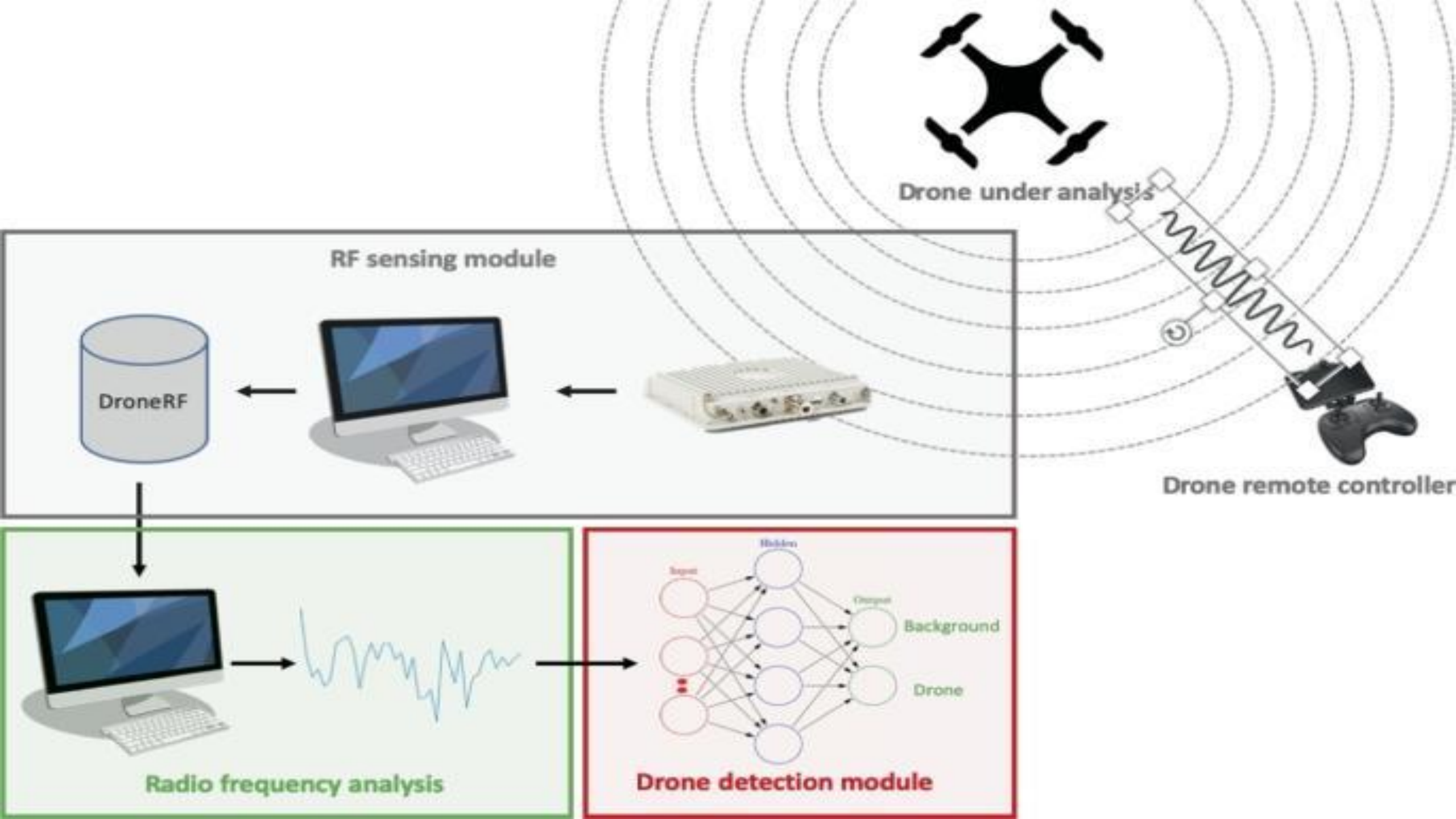
PASSIVER DETECTION UNIT

- Monitors wireless signal activity in the surrounding area
- Drones relyon RF communication with controllers
- High RF activity indicates: Possible drone control signals and Suspicious air activity

Advantage:

Passive detection No signal transmission(stealthy)1q



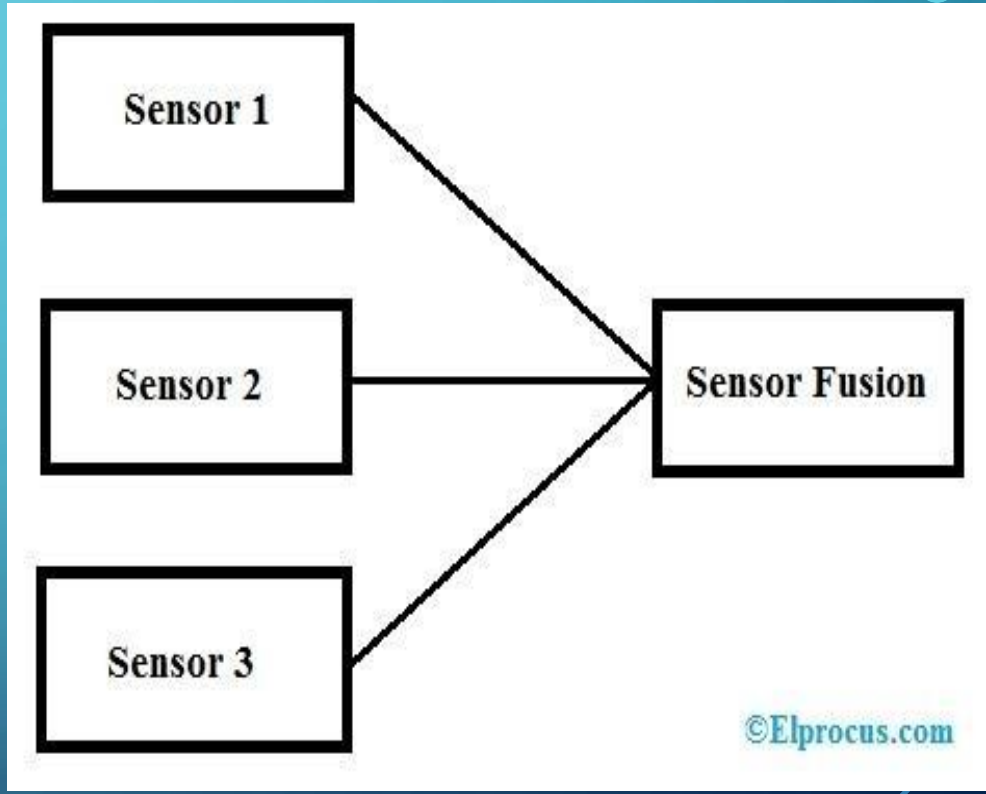

Drone under analysis
RF sensing module
DroneRF
Drone remote controller
Background
Drone
Radio frequency analysis
Drone detection module

EMBBED CONTROL UNIT

- The system is implemented using:
- ArduinoorESP32,based on availability
- Functions of the controller:
- Sensor data acquisition
- Pattern recognition
- Sensor fusion
- Threat classification
- Output control

SENSOR FUSION CONCEPT

- Individual sensors may generate false alarms
- Combining multiple sensors improves accuracy
- **Example:**
- Rfactivity+drone-likesound→Drone conformed
- Sound only→possible treat
- NoRF+bird sound→Safe zone

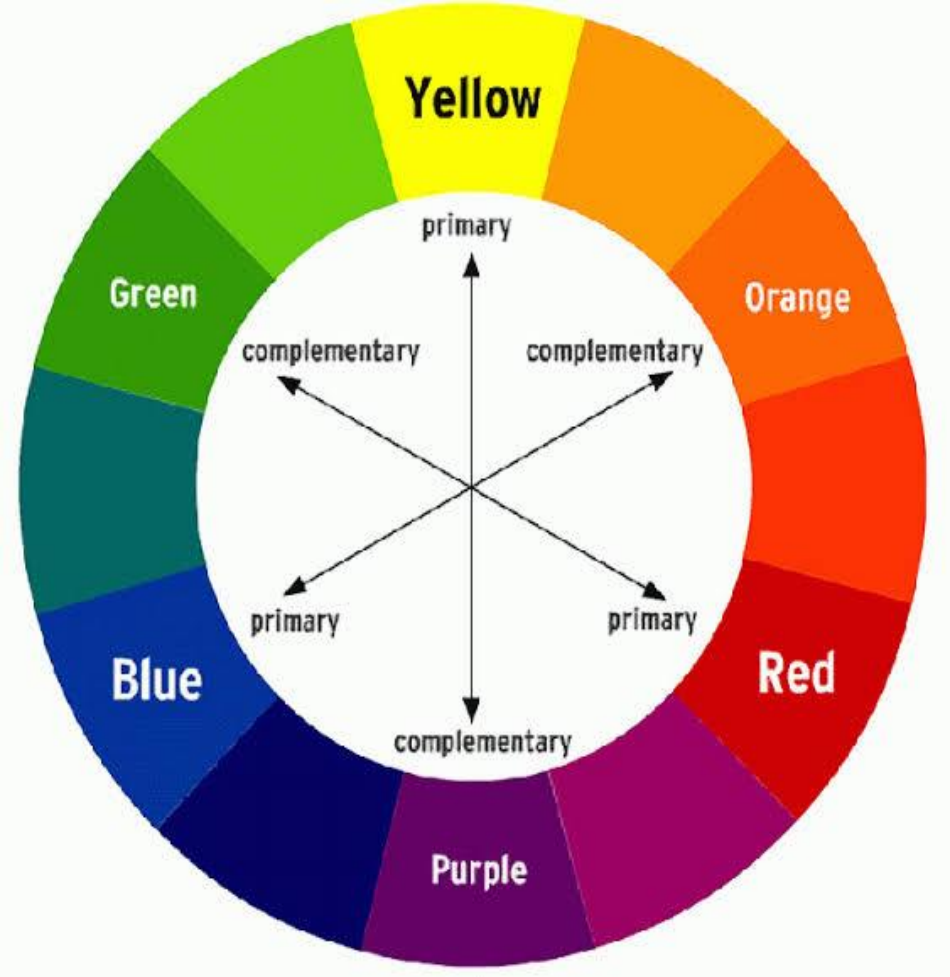


DECISION LOGIC

- Step1:Check ground vibration sensor
- Step2:Check acoustic sensor
- Step3:Check Rf activity
- Step4:Apply classification rules
- Step5:Display threatty peandalert level
- This rule-based logic simulates defence decision systems.l

THREAD LEVEL CLASSIFICATION

- **Low Threat(Green):**
- Bird noise
- NoRFactivity
- **Medium Threat(Yellow):**
- Single sensor trigger
- **High Threat(Red):**
- Confirmed drone
- Vehicle movement

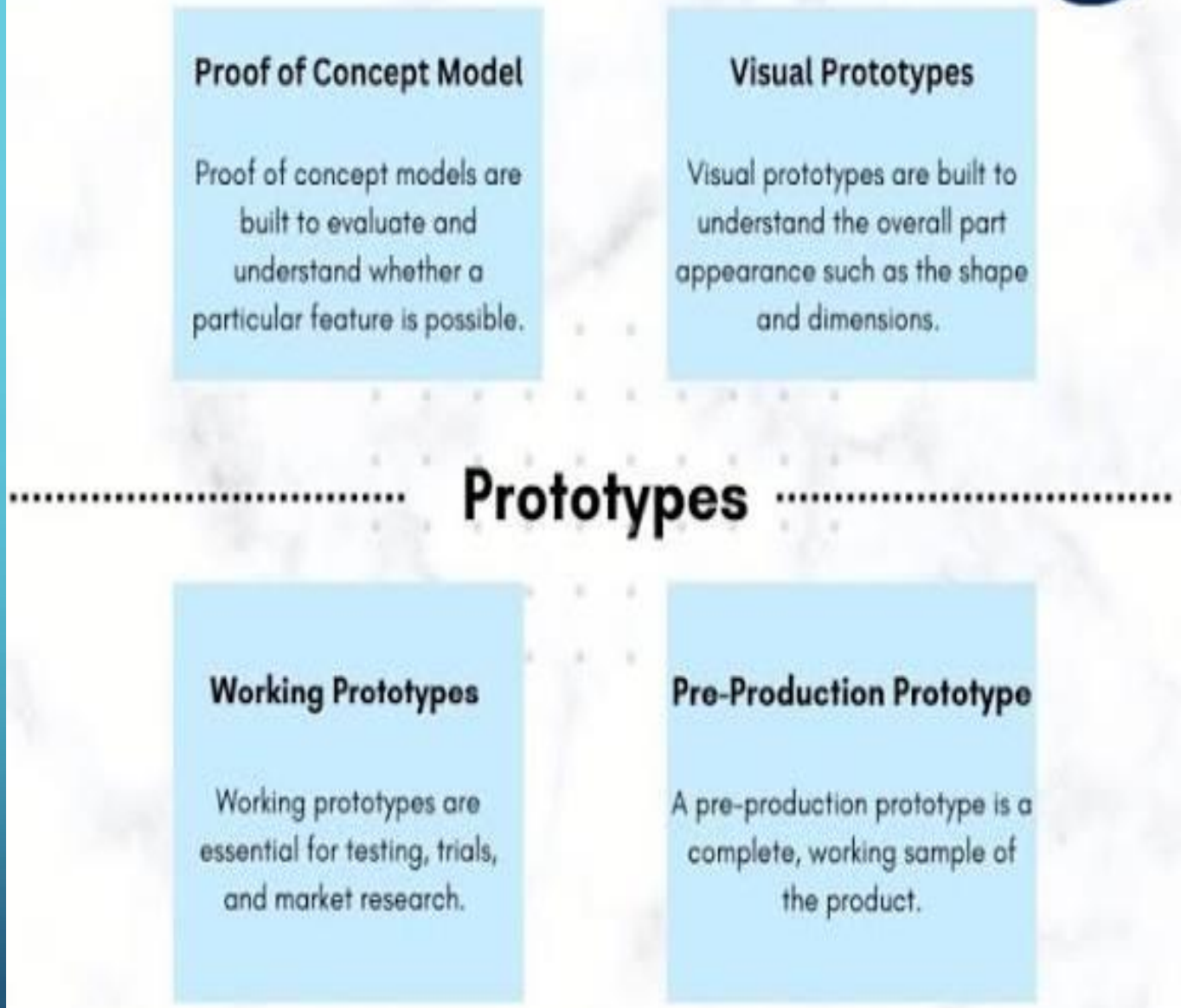


OUTPUT AND ALERT SYSTEM

- LCD Display:
- SAFEZONE
- LAND:HUMAN
- LAND:VEHICLE
- AIR:DRONE CONFIRMED
- LED Indicators:
- Green→Safe
- Yellow→Warning
- Red→Danger
- Buzzer for audible alert

PROTOTYPE MODEL DESIGN

- Ground section contains vibration sensor
- Elevated tower contains acoustic and RFsensors
- Control unit contains controller,LCD,and alerts
- Designed to resemble a border surveillance post



ADVANTAGES

- Low-costand compact
- Reduced false alarms
- Modularandscalable
- Operates in low visibility conditions
- Defence-oriented architec

DISADVANTAGE

- Limited detection range
- Basic sound classification
- Environmental noise may affect accuracy
- Prototype-level implementation

FUTURE SCOPE

- AI based sound and vibration analysis
- Long range communication (LoRa/ GSM)
- Thermal detection for camouflaged intruders
- Direction-based threat detection
- Centralized monitoring dashboard

CONCLUSION

- Multi-sensor approach improves intrusion detection accuracy
- Sensor fusion enables intelligent threat classification
- The prototype demonstrates are a realistic defence surveillance concept
- Suitable for further research and real world expansion

Thank You